

Jay Singh

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EDUCATION

IIT Kharagpur

Kharagpur, India

Bachelor of Technology (B.Tech)

July 2026

- **Relevant Coursework:** Probability and Statistics, Machine Learning Foundations, Applied Mathematics, LDA

WORK EXPERIENCE

AI Intern (Research) | Pibit AI (YC W21) | PPO Offered

Jan – May 2026

- Led **Template Mapper**, the classification core of a **loss-run** API system with **>91%** accuracy, over **850** labels
- **Productionized** the end-to-end extraction, ingestion, preprocessing, prompt optimization, evaluation, and QC
- Launched Document Handler with **OCR** framework for ingestion failure detection, saving **4+** operational days

ML Intern | Pascal AI Labs | PPO Offered

Nov – Dec 2025

- Pioneered DFR-Tracer to interpret restatements in SEC 10-K filings, with **LangGraph** achieving 90% accuracy
- Optimized multi-hop **lineage graph** with Pgvector and fuzzy alignment, boosting accuracy to **86%** from 73%
- Eliminated black-box LLM outputs by **hybrid reranking** with source citation and hallucination rejection loops

Data Science Intern | Innovaccer | PPO Offered

Jun – Aug 2024

- Built a classification model identifying **quality measures/risk modules** from client queries with **85%** accuracy
- Enhanced LangChain-Milvus RAG system over 150 pages, refining chunking and cutting latency by **73%** to 8.1s
- Streamlined a **multi-chain** pipeline routing queries across **5+** classifiers in parallel at **7s** average latency at scale

Research Apprenticeship | Cambridge Judge Business School

May – Sep 2023

- Scaled a **DAG-based agentic** workflow, cutting manual research by **90%** to generate data-driven business ideas
- Designed a shared **agent memory** layer for reasoning tasks cutting tool calls by 44% versus no-memory version
- Refined agent execution through parallel task scheduling and response caching, slicing inference cost and latency

PROJECTS

PortSim-RL | CP-SAT + RL Hybrid for Berth/Crane Scheduling

Apr 2026

- Engineered CP-SAT + **MaskablePPO** scheduler, validated vs. **published optimality bounds** on real port data
- Unified **SimPy** simulator, Gym env, and **CP-SAT** solver on a shared feasibility contract for rail-crane scheduling
- Proved RL repair matches CP-SAT quality at **<5ms** latency vs. 30s re-solves across **8** controllers and **7** scenarios

RealTime-ASR-Engine | Low-Latency Speech Recognition Pipeline

Feb 2026

- Composed a real time ASR pipeline with **Whisper-small** + Silero VAD for STT streaming and silence filtering
- Quantized encoder-decoder weights to INT8 and deployed via CTranslate2 C++ engine, achieving **0.038 RTF**
- Benchmarked on **LibriSpeech**, maintaining **94.6% accuracy** while cutting GPU memory by **62% (910 MB)**

Attention-Based Federated Learning Approach for IDS | CSE IIT Kharagpur

May 2025

- Achieved **80%** zero-shot detection accuracy with a 4-layer transformer classifier, residual MLP, and LayerNorm
- Enriched NSL-KDD dataset with metadata-rich **LLaMA** output for **60,000** entries and contextual vector labels
- Applied **FedAdam** optimizer, stabilizing convergence rates by 25% and improving generalization against attacks

“Attention is All You Need” Paper Replica | Built From Scratch

Jan 2025

- Crafted an **encoder-decoder** Transformer using PyTorch with a 10k vocab size and **BPE-based tokenization**
- Utilized multi-head attention with parallel Q/K/V projections to reproduce the original Transformer system
- Implemented position-wise **FFN** with ReLU activations, **residual connections**, **LayerNorm** on each sublayer

CERTIFICATIONS & SKILLS

- **Certifications:** PyTorch for Deep Learning, Career Essentials on Gen AI, Advanced Learning Algorithms
- **Technologies/Frameworks:** Python, TensorFlow, PyTorch, AWS, Snowflake, FastAPI, LangChain, Django, Docker, SQL, RAG, OpenTelemetry, MLflow, ReAct, Git, GitHub, Hugging Face, DSPy, XGBoost, LightGBM
- **Skills:** Applied ML, Recommender Systems, Statistics, Agentic AI, RLHF, Fine Tuning, LLM Evaluation, OCR, CNN, NLP, Federated Learning, Document Extraction, Feature Engineering, MLOps, Transfer Learning, STT